

L08 Diffusion Models Report

1. Understanding Diffusion

The forward diffusion process refers to the gradual degradation of an image, step by step, through the superposition of small amounts of random noise. In the initial stages, the image remains sharp and fully recognizable. As the steps progress, noise accumulates incrementally, causing the image to become increasingly blurred and its structural integrity diminish. Ultimately, after numerous steps, the image becomes indistinguishable from pure random noise.

According to my lab notebook, I observed that during the early steps, the image undergoes only slight distortion; edges soften, and colors begin to blend into one another. In the intermediate phase, the image's primary structural features begin to vanish, and the degree of pixelation intensifies. By the final stage, the image resembles television static, appearing completely devoid of any recognizable shapes.

- Why do we add noise gradually instead of all at once?

Adding noise incrementally allows the model to learn a series of minute and manageable transformations, rather than having to master a single, massive, and complex discontinuous leap. If the noise were applied all at once, the model would only ever observe a pristine image alongside a completely randomized one—a scenario that would render the learning task exceedingly difficult. By introducing noise in stages, the model learns how to perform denoising in an incremental fashion.

This approach optimizes the learning process by simplifying the task: the model needs only to predict minute amounts of noise, rather than having to reconstruct the entire image from scratch. This helps stabilize the training process, as each step is reduced to a

straightforward prediction problem. In my experiments, I observed that images in the intermediate stages—despite containing noise—still retained a degree of structural information, thereby providing the model with useful learning signals.

- Visualization

During the preliminary stages of the denoising process, the images still appear heavily saturated with noise. Roughly halfway through the process, the coarse outlines and shapes of the images begin to emerge. Once the denoised progress reaches approximately 60% to 70%, most of the image's primary features become clearly visible, with only the finer details remaining to be restored.

- Does this vary by image?

The specific "critical point", the stage at which an image becomes recognizable, varies from one image to another. When processing simple images, they typically become recognizable much earlier, sometimes as early as the 30% to 40% mark of the denoising process. However, when dealing with complex images, the time required to process them can be significantly longer. Based on my observations, images characterized by sharp edges or those containing distinct, isolated objects tend to become recognizable sooner; conversely, images with richer details or more cluttered compositions require a greater number of denoising steps before they become clearly discernible.

2. Model Architecture

- U-Net

The U-Net architecture is particularly effective for diffusion models because its design is tailored to simultaneously capture global context and preserve fine spatial

details, precisely what is required for denoising tasks. During the diffusion process, the model must take a noisy image as input and predict either the noise contained within it or the original, clean image.

Through its unique encoder-decoder architecture, U-Net employs an encoder responsible for compressing the image and learning high-level features, while a decoder is tasked with reconstructing the image to its original resolution. Compared to simpler architectures, U-Net's design is sophisticated enough to capture both global context and spatial information. This point is particularly evident in my Lab, even during the intermediate steps of the denoising process, the model can simultaneously reconstruct both the coarse outlines of objects and increasingly distinct details—a feat that simple CNN architectures struggle to achieve.

- Skip Connections

Skip connections refer to the direct links established between corresponding layers of the encoder and decoder. They directly transmit high-resolution feature maps to the deeper layers of the network. Skip connections are crucial in our model because, during the forward diffusion process, noise gradually degrades the excellent details of the image; consequently, during the denoising process, the model must recover these degraded details. Based on my observations, if skipping connections are omitted, the model's output images tend to appear blurry and indistinct, lacking the requisite structural coherence.

- How the model is conditioned to generate specific images?

In our diffusion models, the principle behind Class Conditioning lies in explicitly injecting label information into the denoising process, thereby enabling the model to

precisely determine—at every step—what type of image it should generate. Each class label is first transformed into a dense vector via an embedding layer—a process analogous to the encoding of time steps. During both the training and sampling phases, the model receives three types of inputs: a noisy image, a time-step embedding, and a class embedding. These embedding vectors are integrated into the network architecturally, through element-wise addition, and injected into the intermediate feature maps within the residual blocks of the U-Net structure. Consequently, at every denoising step, the model is guided not merely to remove noise, but to do so in a manner that ensures the denoised result aligns with the specific structural properties and characteristics of the designated class.

For instance, if the model is conditioned to generate a specific digit or object category—even if its initial input consists solely of pure noise—it can gradually sculpt this noisy input into an image form that corresponds to the features of that particular class. This mechanism enables the model to effectively learn and master a conditional distribution, thereby generating images that are both diverse and strictly adherent to the characteristics of the specified class, rather than simply outputting arbitrary, random results.

3. Training Analysis

- Loss Value

The loss value measures the accuracy with which the model predicts the noise added to an image. Based on my observations, during the initial stages of training, the loss value is relatively high and highly volatile fact consistent with the appearance of the

generated image outputs at that time, which looked entirely like random noise. A higher loss value indicates that the model's predictive performance is poor and that it is unable to perform denoising effectively. As the loss value decreases, the model's ability to reconstruct meaningful structures improves accordingly.

- Why do we need time embedding in diffusion models?

Time embedding tells the model how much noise is present at a given step. It encodes the current timestep into a vector and feeds it into the network. This helps the model adjust its denoising strategy based on the stage. Early steps focus on coarse structure, while later steps refine details. Without it, the model cannot tell where it is in the denoising process.

4. CLIP Evaluation

The CLIP score serves as a metric to quantify the degree of alignment between a generated image and its corresponding text description or class label. A higher score indicates superior semantic alignment between the image and the text, implying that the generated image is more readily recognizable. In my lab, images characterized by sharp outlines and distinct features consistently achieved the highest scores, whereas those that were noisy, blurry, and structurally incoherent received the lowest scores.

Images with simple structures and strong contrast are easy for models to generate. Such images require fewer details and exhibit a high degree of consistency across different samples. Conversely, complex images featuring intricate textures or multiple objects are more difficult to generate, as models often struggle to accurately reconstruct fine textural details during the denoising process.

CLIP scores can serve as a guidance signal during the sampling process. This mechanism facilitates a more precise alignment between the generated images and the target categories or text descriptions. As iterations progress, this approach ultimately yields generated images of greater accuracy and higher quality.

5. Practical Applications

Diffusion models such as the one we have built hold significant value across numerous practical applications, for example, generating images for the design and media industries, or enhancing image quality through denoising and super-resolution techniques.

However, our current model still faces several limitations: it is computationally intensive, performs sub optimally when processing extraordinarily complex or detail-rich images, and relies heavily on the quality and diversity of its training data.

If I were to continue developing this project, I would train the model for a longer duration on a larger and more diverse dataset to improve both the quality of image generation and the model's generalization capabilities. I would also optimize the model's architecture to better capture fine-grained details within images.